

ROS 2 Jazzy cheat sheet (ASAP basics) ASAP · ROS 2 Jazzy

Every new terminal

```
source /opt/ros/jazzy/setup.bash           # ROS itself           (in ~/.bashrc after setup)
source ~/asap-kit/scripts/group_env.sh 3    # your group: 3 -> ROS_DOMAIN_ID=13    (in ~/.bashrc after setup)
source ~/ros2_ws/install/setup.bash        # YOUR packages: after every build, in every terminal
```

New terminal: `Ctrl+Alt+T`. New tab: `Ctrl+Shift+T`. Stop a node: `Ctrl+C`.

Workspace → packages → nodes

```
~/ros2_ws/           the workspace: build here, never inside src/
├─ src/              source code: one folder per package
│   └─ counter_pkg/  a package (a unit you build, share and depend on)
│       └─ package.xml  name + dependencies (<depend>std_msgs</depend>)
│           └─ setup.py  entry_points: which nodes `ros2 run` can find
│               └─ setup.cfg  leave alone
│                   └─ resource/  marker file, leave alone
│                       └─ counter_pkg/  your Python code: one file per node
└─ asap_interfaces/  an interface package: msg/ srv/, CMakeLists.txt
└─ build/  install/  log/  made by colcon; safe to delete
```

Create

```
mkdir -p ~/ros2_ws/src && cd ~/ros2_ws/src

# Python package with a first node already wired up:
ros2 pkg create counter_pkg --build-type ament_python --license Apache-2.0 \
  --dependencies rclpy --node-name counter

# Interface package (.msg/.srv need CMake):
ros2 pkg create asap_interfaces --build-type ament_cmake --license Apache-2.0

# Someone else's package: clone it into src/, then build
git clone https://github.com/praiseorji4/dht11_ros.git
```

Package name **first**: `--dependencies` takes every word after it.

Build and run

```
cd ~/ros2_ws
colcon build --symlink-install           # everything
colcon build --symlink-install --packages-select counter_pkg # one package
source install/setup.bash                # then, in every terminal

ros2 run counter_pkg counter              # ros2 run <package> <executable>
ros2 run counter_pkg counter --ros-args -r __node:=counter2 # a second copy with another name
ros2 run counter_pkg counter --ros-args -p name:=value      # set a parameter at start
ros2 launch dht11_ros dht11.launch.py mode:=sim             # start nodes from a launch file
```

You changed	Rebuild?
a .py node (with --symlink-install)	no, just restart the node
setup.py, package.xml, a new node file	yes
a .msg / .srv / CMakeLists.txt	yes, then source install/setup.bash

Inspect

node	<code>ros2 node list</code> · <code>ros2 node info /counter</code>
topic	<code>ros2 topic list -t</code> · <code>echo /count</code> · <code>hz /count</code> · <code>info /count</code> <code>ros2 topic pub --once /count std_msgs/msg/Int32 "{data: 5}" (-r 1 = once a second)</code>
service	<code>ros2 service list -t</code> · <code>type /spawn</code> <code>ros2 service call /spawn turtlesim/srv/Spawn "{x: 2.0, y: 2.0, name: turtle2}"</code>
param	<code>ros2 param list /turtlesim</code> · <code>get /turtlesim background_r</code> · <code>set /turtlesim background_r 150</code>
action	<code>ros2 action list -t</code> · <code>ros2 action send_goal --feedback <name> <type> "{...}"</code>
interface	<code>ros2 interface show std_msgs/msg/Int32</code> · <code>ros2 interface package asap_interfaces</code>
pkg	<code>ros2 pkg list</code> · <code>ros2 pkg executables counter_pkg</code>
picture	<code>rqt_graph</code> (press the refresh button after starting nodes)

`ros2 param set` is strict about types: a double parameter needs `25.0`, not `25`.

The four ways nodes talk

	Use when	Shape	Example today
Topic	a stream of data or state	publisher → many subscribers, no reply	<code>/count</code> , <code>/temperature</code>
Service	a short question with a quick answer	one request → one response	<code>/spawn</code> , <code>/validate_reading("too hot?")</code>
Action	a long job with progress and cancel	goal → feedback... → result	<code>/turtle1/rotate_absolute</code>
Parameter	a setting you can change while it runs	get / set by name	<code>background_r</code> , <code>max_temp</code>

Snippets

`setup.py`, one line per node. Mind the **comma** between entries:

```
'console_scripts': [  
    'counter = counter_pkg.counter:main',  
    'count_listener = counter_pkg.count_listener:main',  
],
```

Interface package `CMakeLists.txt`, after `find_package(ament_cmake REQUIRED)`:

```
find_package(builtin_interfaces REQUIRED)  
find_package(rosidl_default_generators REQUIRED)  
rosidl_generate_interfaces(${PROJECT_NAME}  
    "msg/Count.msg"  
    DEPENDENCIES builtin_interfaces  
)
```

Interface package `package.xml`, the first three lines under `<buildtool_depend>ament_cmake</buildtool_depend>`, the last one just above `<export>`:

```
<buildtool_depend>rosidl_default_generators</buildtool_depend>  
<depend>builtin_interfaces</depend>  
<exec_depend>rosidl_default_runtime</exec_depend>  
  
<member_of_group>rosidl_interface_packages</member_of_group>
```

Stuck?

```
bash ~/asap-kit/scripts/verify.sh 3      # is step 3 working? PASS/FAIL per check  
bash ~/asap-kit/scripts/catchup.sh 3    # give me the finished step 3 (your src/ is backed up)
```